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Institution: Federal University of Petroleum Resources, Effurun
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Levels: 400

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Paper 1 of 3 — PEE 410 APPLIED RESERVOIR ENGINEERING PQ 2025

Level 400 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
FIRST SEMESTER EXAMINATION, 2024-2025 SESSION**

PNG 410: APPLIED RESERVOIR ENGINEERING
Date: 13-04-2025 TIME ALLOWED: 3HR UNIT: 3

**INSTRUCTION: ANSWER THREE QUESTION IN SESSION A AND
TWO IN SESSION B**

QUESTION 1**SESSION A**

An active water drive reservoir has a cumulative water production of 28640 bbl was estimated after the first year of production. The pressure profile, reservoir and aquifer properties are given as:

Time (yrs)	0	1	2	3	4	5
Pressure (psia)	5027	5016	5014	4966	4818	4775
W_{ep}		5.3168	8.7385	11.7766	14.6216	17.3312

Aquifer permeability, K_w	160 mD
Water viscosity, μ_w	0.78 cp
Drainage radius, r_e	9500 ft
Aquifer radius, r_a	47500 ft
Total compressibility, C_t	$3.80 \times 10^{-6} \text{ psi}^{-1}$
Porosity, ϕ	22%
Aquifer angle	120°

Calculate cumulative water influx for each year

QUESTION 2

The result of a gas chromatographic test performed at FUPRE Laboratory on a gas sample collected from a gas field in the Niger Delta region is shown in the table below

Components	Mole Percent (xi)
Nitrogen	0.18
Carbon dioxide	0.91
Methane	87.43
Ethane	4.41
Propane	3.65
i-butane	0.64
n-butane	1.50
i-pentane	0.45
n-pentane	0.38
Hexane	0.30
Heptane Plus	0.15

Additional information:

Initial reservoir pressure, P_i	3200 psia
Reservoir thickness, h	55 ft
Connate water saturation, S_{wc}	23%
Area, A	110 acres
Porosity, ϕ	19.8%
Reservoir temperature	200°F

- Calculate the cumulative gas produced at 2500 psia
- Calculate the recovery factor at 2500 psia

- Ultimate oil recovery (N_{ul}) to economic limit
- Production rate (q_{os}) at end of year 2008

Solution

The effective rate is:

$$D_i' = \frac{q_i - q_{i+1}}{q_i} = \frac{973 - 827}{973} = 0.1501/\text{yr} = 15.01\%/\text{yr}$$

Nominal decline rate is:

$$e^{-D_i} = 1 - D_i'$$

$$D_i = -\ln(1 - D_i') = \ln(1 - 0.1501) = 0.1626/\text{yr} = 16.26\%/\text{yr}$$

The remaining reserve:

$$N_r = \frac{q_i - q_a}{D_i} = \frac{q_i - q_{EL}}{D_i} = \frac{(973 - 190) \text{ stb/month} * 12 \text{ months/yr}}{0.1626/\text{yr}} = 57785.97 \text{ stb}$$

Abandonment time (time to economic limit)

$$t_a = \frac{1}{D_i} \ln\left(\frac{q_i}{q_{EL}}\right) = \frac{1}{0.1626} \ln\left(\frac{973}{190}\right) = 10.05 \text{ yrs}$$

The ultimate oil recovery to economic limit is

$$N_{ul} = N_p + N_r = 80519 + 57785.97 = 138304.97 \text{ stb}$$

The production rate at the end of 2008 is

$$q_{os} = q_{04} e^{-D_i t}$$

$$q_{04} = 827 \text{ stb/month}$$

Hint:

$$q_{os} = 827 e^{-0.1626 \cdot 12} = 117.52 \text{ stb/month}$$

$$\frac{2.309kt}{\mu_w \phi_w c_{rw} r^2} = 1.119 f \phi h c_{rw} r^2$$

$$\frac{\theta \mu C_r r^2}{4(0.000264)kt} = P_i + 70.6 \frac{q \beta \mu}{kh} E_i \left\{ -\frac{\theta \mu C_r r^2}{0.001056kt} \right\}$$

$$\frac{162.6 q \beta \mu}{kh} \left[\log t + \log \left\{ \frac{k}{\theta \mu C_r r^2} \right\} - 3.23 + 0.8685s \right]$$

$$\Delta V_b = \frac{h}{2} (A_n + A_{n+1}) = \sum_{i=1}^n \Delta V_{bi}$$

$$\Delta V_b = \frac{h}{3} (A_n + A_{n+1} + \sqrt{A_n * A_{n+1}}) = \sum_{i=1}^n \Delta V_{bi}$$

$$\frac{q_i}{(1 + 0.5 D_i t)^{1/2}} = \frac{1}{D_i} = \left(\frac{q_i}{q_a} - 1 \right) = \frac{q_i}{(1 - b) D_i} \left\{ 1 - \left(\frac{q_a}{q_i} \right)^{1-b} \right\}$$

$$\frac{1}{t} \ln \frac{q_i}{q_a} = \frac{q_i}{D_i} \ln(1 + D_i t)$$

QUESTION 3

1) You are provided with the following data:

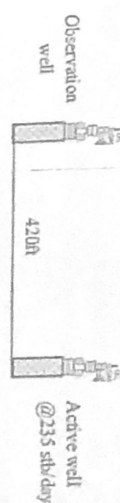
Area of the field	5400 acres
Thickness of the reservoir	37 ft
Porosity of the formation	20.1% for the top 12ft 19.4% for the middle 17ft 17.6% for the bottom 8ft
Water saturation	26.3% for the top 12ft 22.5% for the middle 17ft 18.9% for the bottom 8ft
Gas oil ratio	500 cuft/bbl
Net reservoir sand thickness	37ft
Temperature	200°F
Gravity	0.68
Gross reservoir sand thickness	35° API
	40ft

- $B_o = 0.972 + 0.000147 \left[R_s \left(\frac{Y_o}{Y_o} \right)^{0.5} + 1.257 \right]^{1.175}$
- Calculate the oil initially in place
 - Calculate the stock tank oil initially in place
 - Calculate the STOIP if a cut-off value of 10% is applied to the porosity and 15% to the water saturation.

- (ii) Use one diagram to show the variation of R_s , μ_o , B_g and B_o with pressure

QUESTION 4

- State four factors affecting the productivity index
- The figure below shows an active and an observation well. Calculate the pressure change in the observation well at:
 - A distance of from the active well after 10 hrs of production
 - The wellbore after 10 hrs of production



Wellbore radius, r_w	0.48 ft
Total compressibility, C_t	$18.9 \times 10^{-6} \text{ psi}^{-1}$
Reservoir height, h	35 ft
Initial pressure, P_i	3860 psi
Formation permeability, k	997 mD
Oil viscosity, μ_o	0.86 cp
Oil formation volume factor, β_o	1.132 rb/stb
Porosity, ϕ	20.7%

QUESTION 5

The result Uduia gas field volumetric analysis and production data is given below

Parameter	Value
ultimate recoverable gas reserves (MMscf)	37.82
restricted production rate (MMscf/month)	385
nominal decline rate (per month)	0.035
economic limit (MMscf/month)	23

SESSION B

Calculate the following, assuming exponential decline

- The life of the well,
- The prorated time

- iii. Cumulative gas production during the restricted rate
 iv. The cumulative production the year production starts declining

QUESTION 6

- (a) Calculate the average reservoir pressure at the Datum depth of 8750 ftss for the following fluid pressure gradients, given that the GOC and OWC are at 8700 ftss and 8800 ftss respectively:

$$\left(\frac{dp}{dd}\right)_{\text{gas}} = 0.08 \text{ psi/ft}, \left(\frac{dp}{dd}\right)_{\text{oil}} = 0.269 \text{ psi/ft}, \left(\frac{dp}{dd}\right)_{\text{water}} = 0.434 \text{ psi/ft}$$

Well A = 3685 psig at 8690 ftss
 Well B = 3716 psig at 8800 ftss
 Well C = 3725 psig at 8820 ftss
 Well D = 3689 psig at 8710 ftss
 Well E = 3713 psig at 8790 ftss

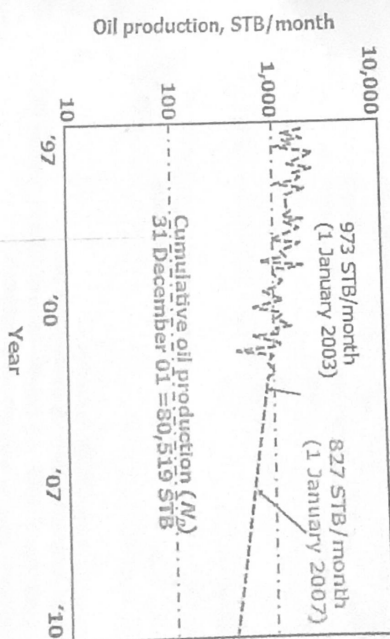
- (b) A well penetrates a reservoir near the top of a fluid column. The GOC has been detected by logs but the OWC was not. An oil sample was taken at 7890 ft TVD with a pore pressure of 3890 psig recorded. The field water gradient is 0.445 psi/ft, oil gradient is 0.315 psi/ft. find the OWC.

QUESTION 7

Analyze the C4 reservoir with production and fluid data given in the table below. Based on your findings, recommend the C4 reservoir management which secondary recovery technique to adopt to further recover the oil.

Initial reservoir pressure = 2740 psi	Current reservoir pressure = 2460 psi
Initial oil FVF = 1.3985 rb/stb	Initial gas exp. Factor = 198.3 scf/cuft
Initial solution GOR = 643 scf/stb	gas exp. Factor = 178.14 scf/cuft
oil FVF = 1.3578 rb/stb	Cum. Gas Produced = 15498 MMscf
Current solution GOR = 577.3 scf/stb	Gas/oil sand volume ratio = 0.7
Cum. Oil Produced = 18.9 MMstb	Cum. Water produced = 3.3 MMstb
Connate water saturation = 17%	Formation compressibility = $3.5 \times 10^{-6} \text{ psi}^{-1}$
STOIIP = 120 MMstb	Water compressibility = $3.5 \times 10^{-6} \text{ psi}^{-1}$

The production trend of Okoso oil field is presented in the figure below. The estimated economic limit of this field is 190 stb/month.



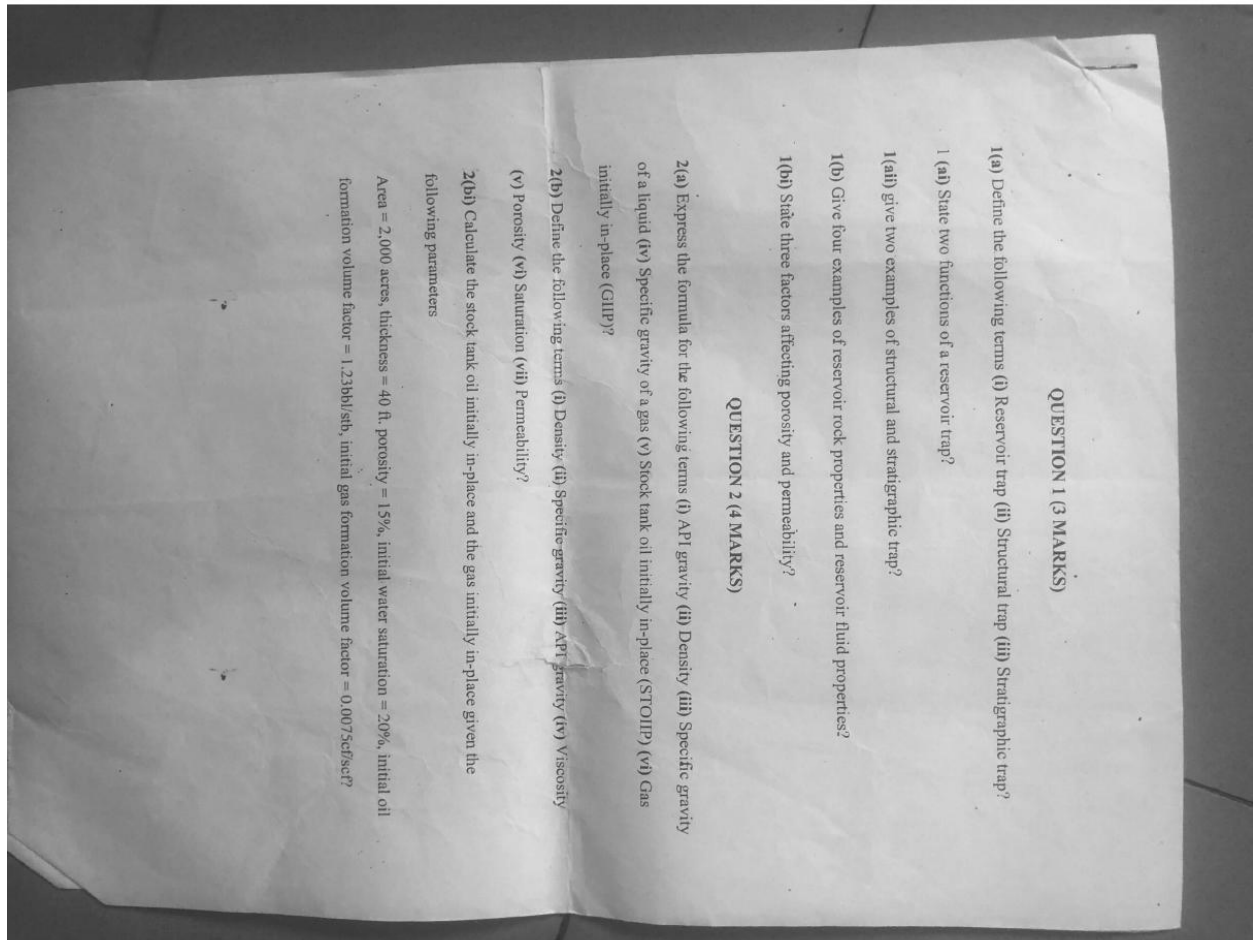
Calculate

- Effective (D') and nominal (D) decline rates
- Remaining reserves (N_p) from 1 January 2003 to the field's economic limit
- Time (t) to produce to economic limit

Paper 2 of 3 — PEE 410 Reservoir Engineering

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QUESTION 3 (3 MARKS)

- 3(a) Define the following terms (i) Wet gas (ii) Dry gas (iii) Associated gas (iv) Non-associated gas (v) Condensate gas?
- 3(aii) State the API gravity range for (i) Light crude oil (ii) Medium crude oil (iii) Heavy crude oil?
- 3(b) Petroleum in solid form, liquid form and gaseous form are called?
- 3(bi) State two functions of upstream sector?
- 3(bii) Define the term exploration and production?

Paper 3 of 3 — PEE 410 (RESERVOIR GAS AND OIL PROPERTIES)

Level 400 • First Semester • Handout

Chapter 1 Review: Reservoir, Gas and Oil Properties

1.1 Introduction to Reservoir Engineering

Crude oil, natural gas, and water are the substances that are of chief concern to petroleum engineers. Although these substances sometimes occur as solids or semisolids, usually at lower temperatures and pressures, as paraffin, gas- hydrates, ices, or high pour-point crudes, in the ground and in the wells they occur mainly as fluids, either in the vapor (gaseous) or in the liquid phase or, quite commonly, both. Even where solid materials are used, as in drilling, cementing, and fracturing, they are handled as fluids or slurries. The division of the well and reservoir fluids between the liquid and vapor phases depends mainly on the temperature and pressure. The state or phase of a fluid in the reservoir usually changes with pressure, the temperature remaining substantially constant. In many cases the state or phase in the reservoir is quite unrelated to the state of the fluid when it is produced at the surface. The precise knowledge of the behavior of crude oil, natural gas, and water, singly or in combination under static conditions or in motion in the reservoir rock and in pipes and under changing temperature and pressure, is the main concern of petroleum engineers.

In 1928: serious consideration to gas-energy relations. Sclater and Stephenson describe the first recording bottom hole pressure gauge and thief for sampling fluids under pressure in wells.

In 1933: Fancher, Lewis, and Barnes made one of the earliest petrophysical studies of reservoir rocks.

In 1934: Wycoff, Botset, Muskat, and Reed developed a method for measuring the permeability of reservoir rock samples based on the fluid flow equation discovered by Darcy in 1856.

In 1935: Schilthuis described a bottom-hole sampler and a method of measuring the physical properties of the samples obtained.

In 1940: Leverett and Lewis reported research on the three phase flow of oil, gas, and water.

During 1960s: the terms reservoir simulation and reservoir mathematical modeling became popular.

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With the development of petroleum techniques (finite-difference or finite-element techniques), concepts, and equations, reservoir engineering became a powerful and well defined branch of petroleum engineering.

1.1.1 Critical Concepts

Petroleum Engineering may be defined as the application of scientific principles to the drainage problems arising during the development and production of oil and gas reservoirs. It has also been defined as “the art of developing and producing oil and gas fluids in such a manner as to obtain a high economic recovery.”

The working tools of the reservoir engineer are:

- Subsurface geology,
- Applied mathematics,
- The basic laws of physics and chemistry (Governing the behavior of liquid and vapor phases of crude oil, natural gas, and water in reservoir rocks.)

Oil and gas accumulations occur in underground traps formed by structural and/or stratigraphic features. Fortunately they usually occur in the more porous and permeable portions of beds, which are mainly sands, sandstones, limestones, and dolomites, in the intergranular openings, or in pore spaces caused by joints, fractures, and solution activity. A reservoir is that portion of a trap which contains oil and/or gas as a single hydraulically connected system. Many hydrocarbon reservoirs are hydraulically connected to various volumes of water-bearing rock called aquifers.

The initial production of hydrocarbons from the underground reservoir is accomplished by the use of natural reservoir energy and is referred to as primary production. The oil and gas are displaced to production wells under primary production by:

- Fluid expansion□ no aquifer and no fluid is injected into the reservoir
- Fluid displacement□inject gas
- Gravitational drainage□water influx

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- Capillary expulsion□water influx

Secondary recovery operation is the use of either a natural gas or a water injection scheme. The process of secondary recovery is waterflooding.

- The main purpose of either a natural gas or a water injection process is to maintain the reservoir pressure.

Tertiary recovery processes is to inject the materials which not normally present in reservoir, excludes conventional water flooding.

Figure 1.1. Oil recovery methods. (Source: Adapted from the Oil & Gas Journal, Apr. 23, 1990)

1.1.2 PVT Diagram and Reservoir Types

a. Phase Behavior

- Phase: A homogenous region of a material system with uniform physical and chemical characteristics.
- Phase behavior: The manner in which hydrocarbons behavior when pressure and temperature change.
- Factors influencing the physical behavior of molecules:
 - Temperature: A physical measure of the average kinetic energy of the molecules of a material. A reflection of the kinetic energy of the molecules.
 - Pressure: a measure of the number of times the molecules of a phase strikes the

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walls of its container. A reflection of the number of molecules present and their

motions.

□ Molecular attraction and repulsion: the forces of attraction and repulsion between molecules.

b. Phase Diagram for Single Component

Phase diagram is not of interest, except for illustration purposes.

Figure 1.2. Phase diagram for Single Component

c. Pressure-Temperature Phase diagrams for Multi-Component System

The conditions under which these phases exist are a matter of considerable practical importance. Figure 1.3 shows a typical pressure-temperature diagram of a multi-component system with a specific overall composition. Although a different hydrocarbon system would have a different phase diagram, the general configuration is similar.

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Figure 1.3. Typical p-T diagram for a multicomponent system.

□ Key Points and Definition:

To fully understand the significance of the pressure-temperature diagrams, it is necessary to identify and define the following key points on these diagrams:

□ Cricondetherm (Tct): the Cricondetherm is defined as the maximum temperature above which liquid cannot be formed regardless of pressure (point E). The corresponding pressure is termed the Cricondetherm pressure.

□ Cricondenbar (pcb): the Cricondenbar is the maximum pressure above which no gas can be formed regardless of temperature (point D). The corresponding temperature is called the Cricondenbar temperature Tcb (problematic if using the above diagram).

□ Critical point: the critical point for a multicomponent mixture is referred to as the state of pressure and temperature at which all intensive properties of the gas and liquid phases are equal (point C). At the critical point, the corresponding pressure and temperature are called the critical pressure P_c and critical temperature T_c of the mixture.

□ Phase envelope (two-phase region): the region enclosed by the bubble-point

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curve and the dew-point curve (line BCA), wherein gas and liquid coexist in equilibrium, is identified as the phase envelope of the hydrocarbon system.

□ Quality lines: the dashed lines within the phase diagram are called quality lines. They describe the pressure and temperature conditions for equal volumes of liquids. Note that the quality lines converge at the critical point (point C).

□ Bubble-point curve (line BC) is defined as the line separating the liquid-phase region from the two-phase region.

□ Dew-point curve (line AC) is defined as the line separating the vapor-phase

region from the two-phase region.

- Uses:
- Classify reservoirs
- Classify the naturally occurring hydrocarbon systems
- Describe the phase behavior of the reservoir fluid

1.1.3 Reservoir Classification

In general, reservoirs are classified on the basis of the location of the point representing the initial reservoir pressure P_i and temperature T with respect to the pressure-temperature diagram of the reservoir fluid. Accordingly, reservoirs can be classified into basically two types:

- Oil reservoirs: if the reservoir temperature T is less than the critical temperature T_c of the reservoir fluid, the reservoir is classified as an oil reservoir.

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- Gas reservoirs: if the reservoir temperature is greater than the critical temperature of the hydrocarbon fluid, the reservoir is considered a gas reservoir.

These broad classifications are further subdivided into five types of Reservoir Petroleum Fluids, depending on:

- The composition of the reservoir hydrocarbon mixture.
- Initial reservoir pressure and temperature
- Pressure and temperature of the surface production

a. Five Types of Reservoir Fluids

- Black Oil (Figure 1.4), also called low shrinkage crude oil or ordinary oil.
- Volatile Oil (Figure 1.5): also called high-shrinkage crude oils. Volatile oils contain relatively fewer heavy molecules and more intermediates (defined as ethane through hexanes) than black oils.
- Retrograde Gas (Figure 1.6): also called retrograde gas&-condensates, retrograde condensate gases, gas condensates, or condensates.
- Wet Gas (Figure 1.7): the entire phase diagram of a hydrocarbon mixture of predominately smaller molecules will lie below reservoir temperature. A wet gas exists solely as a gas in the reservoir throughout the reduction in reservoir pressure. The pressure path does not enter the phase envelope. Thus, no liquid is formed in the reservoir. However, separator conditions lie within the phase envelope, causing some liquid to be formed at the surface.
- Dry Gas (Figure 1.8): thy gas is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and that normal surface separator conditions fall outside the phase envelope. Thus, no liquid is formed at the surface.

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Figure 1.4. Phase diagram of a typical black oil with line of isothermal reduction of reservoir pressure,

1□2□3, and surface separator conditions.

Figure 1.5. Phase diagram of a typical volatile oil with line of isothermal reduction of reservoir pressure,

1-2-3, and surface separator conditions.

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Figure 1.6. Phase diagram of a typical retrograde gas with line of isothermal reduction of reservoir pressure,

1-2-3, and surface separator conditions.

Figure 1.7. Phase diagram of a typical wet gas with line of isothermal reduction of reservoir pressure,

1-2, and surface separator conditions.

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Figure 1.8. Phase diagram of a typical dry gas with line of isothermal reduction of reservoir pressure,

1-2, and surface conditions.

1.2 Review of reservoir, gas and oil properties

1.2.1 Review of rock properties

a. Porosity

The porosity (ϕ) of a porous medium is defined as the ratio of void space, or pore volume, to the total bulk volume of the rock.

$$\phi = \frac{V_p}{V_b} \quad (1.1)$$

This ratio is expressed either as a fraction or in percent. When using a value of porosity in an equation it is nearly always expressed as a fraction.

Hydrocarbon porosity: part of the porosity that contains hydrocarbon. It is the total porosity multiplied by the fraction of the pore volume that contains hydrocarbon.

Total porosity: represents the ratio of the volume of all pores to the bulk volume of a material, regardless of whether or not the pores are interconnected.

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Effective porosity: it is the ratio of the interconnected pore volume to the bulk volume of a material. This is the type of porosity usually measured in the laboratory and used in calculations of fluid flow.

The laboratory methods of measuring porosity include Boyle's law, water-saturation, and organic-liquid saturation methods.

□ Boyle's Law Method

A known volume of helium gas at a fixed initial pressure is isothermally expanded into an unknown void volume. After expansion the resultant stabilized pressure is measured; that pressure being dependent on the magnitude of the unknown volume. The magnitude of the unknown volume may be calculated using the Ideal Gas equation of state.

Boyle's law (constant temperature):

$$P_1 V_1 = P_2 V_2 \quad (1.2)$$

$$\frac{V_2}{V_1} = \frac{P_1}{P_2} \quad (1.3)$$

where,

P_1 = Initial Absolute Pressure, psi

V_1 = Initial Volume, ft³

P_2 = Expanded Absolute Pressure, psi

V_2 = Expanded Volume, ft³

□ Liquid Saturation Method

A fluid of known density is introduced into a dry core sample that is vacuumed. The difference between the weight of the saturated sample (W_{sat}) and the dry weight of the sample (W_{dry}) is the fluid weight that filled in the pore space of the core. If the density of the introduced fluid is known, the pore volume can be determined.

$$\rho = \frac{W}{V} \quad (1.4)$$

$$V = \frac{W}{\rho} \quad (1.5)$$

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where,

W_{fluid} = Weight of fluid, g

W_{sat} = Weight of saturated sample, g

W_{dry} = Dry weight of the sample, g

V_{pore} = Pore volume, m³ Figure 1.9. Schematic Diagram of Vacuum

3 Pump Systems

V_{fluid} = Fluid volume, m

ρ_{fluid} = Fluid density, g/m³

b. Isothermal compressibility

The isothermal compressibility (C) for a substance is given by the following equation:

$$C = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (1.6)$$

where,

C = Isothermal compressibility, psi⁻¹

V = Volume, ft³

P = Pressure, psi

The equation describes the change in volume that a substance undergoes during a change in pressure while the temperature is held constant. When the internal fluid pressure within the pore spaces of a rock, which is subjected to a constant external (rock or overburden) pressure, is reduced, the bulk volume of the rock decreases while the volume of the solid rock material (e.g., the sand grains of a sandstone) increases. Both of these volume changes act to reduce the porosity of the rock slightly.

However, the change in pore volume is nonlinear and the pore volume compressibility is not constant. Newman measured isothermal compressibility and porosity values in 79 samples of consolidated sandstones under hydrostatic pressure. When he fit the data to hyperbolic equation, he obtained the following correlation:

$$C = \frac{C_0}{1 + P/P_0} \quad (1.7)$$

+ . $\emptyset$.

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c. Fluid saturations

The saturation (S) of a fluid is the ratio of the volume that a fluid occupies to the pore volume.

The saturations of all fluids present in a porous medium add to 1.

There are two ways of measuring original fluid saturations: the direct approach and the indirect approach.

□ The Direct Approach

The direct approach involves either the extraction of the reservoir fluids or the leaching of the fluids from a sample of the reservoir rock, which include retorting the fluids from the rock, distilling the fluids with a modified ASTM (American Society for Testing and Materials) procedure, and centrifuging the fluids. Each method relies on some procedure to remove the rock sample from the reservoir. Experience has found that it is difficult to remove the sample without altering the state of the fluids and/or rock.

□ The Indirect Approach

The indirect approach relies on a measurement of some other property, such as capillary pressure, and the derivation of a mathematical relationship between the measured property and saturation.

1.2.2 Review of oil properties

Properties of crude oil and natural gas are fundamental for designing and analyzing oil and gas production systems in petroleum engineering.

Oil properties include:

- solution gas oil ratio
- oil formation volume factor
- oil isothermal compressibility
- oil viscosity

a. Solution gas oil ratio

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A crude oil is said to be saturated with gas at any pressure and temperature if on a slight reduction in pressure some gas is released from solution. Conversely, if no gas is released from solution, the crude oil is said to be undersaturated at the pressure. The undersaturated state implies that there is a deficiency of gas present and that had there been an abundance of gas present, the oil would be saturated at that pressure. The undersaturated state further implies that Figure 1.10. Schematic of typical P-T diagram.

there is no free gas in contact with the crude oil (i.e., there is no gas cap).

Solution gas oil ratio (R_{so}): the volume of gas that can be dissolved in unit volume of oil in reservoir condition.

$$= (1.8)$$

where,

R_{so} = Solution gas oil ratio (GOR), SCF/STB

V_{gas} = Gas volume in standard condition, SCF

Figure 1.11. Solution gas oil ratio.

V_{oil} = Oil volume in stock tank condition, STB

- GOR remains constant when $P > P_b$ (bubble point pressure).
- It decreases as pressure decrease when $P < P_b$.
- It can be measured in PVT laboratories or can be estimated by experimental correlations.

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Figure 1.12. Solution gas oil ratio of the Big Sandy Field reservoir oil, by flash liberation at reservoir temperature of 160°F.

- Solution gas = 567 SCF/STB @ $P_i=3500$ psi
- No gas is evolved from solution when the pressure drops from the initial pressure at 2500 psia (bubble point pressure)
- At 1200 psi, the solution gas is 337 SCF/STB,

One widely used correlation to estimate it is:

$$= \diamond \diamond (1.9)$$

where,

$$\diamond = 0.00091T - 0.0125P_o, \text{ API}$$

T = Temperature, °F

P = Pressure, psia

Solution GOR is used for:

- Volumetric Oil & Gas calculation.
- To estimate other properties of fluid like density.
- Initial GOR can be an indicator of reservoir fluid type.

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b. Oil formation volume factor

The oil formation volume factor (B_o) is defined as the volume of oil in reservoir over volume of oil in surface.

$$= (1.10)$$

where,

B_o = Oil formation volume factor, RVB/STB

V_{res} = Oil volume in reservoir condition, RVB

Figure 1.13. Oil formation volume

V_{st} = Oil volume in stock tank condition, STB factor.

- B_o is always greater than 1 because: in reservoir condition oil can dissolve more gas than in standard condition.
- When $P > P_b$, B_o decreases slowly as pressure increases.

□ When $P < P_b$, B_o decrease as pressure decreases.

Figure 1.14. Formation volume factor of the Big Sandy Field reservoir oil, by flash liberation at reservoir temperature of 160 °F.

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Figure 1.15. Visual conception of the change in single phase and in two-phase formation volume factors for the Big Sandy reservoir fluid.

The single-phase formation volume factor may be estimated from the solution gas, the gravity of the solution gas, the API gravity of the tank oil, and reservoir temperature using a correlation prepared by Standing. Beggs presents Standing's correlation for the oil formation volume factor in equation form as:

□ For $P \leq P_b$:

$$B_o = 1.0 + 0.0001 \left(\frac{P_b}{T} \right)^{1.75} \quad (1.10)$$

where,

$$F = \frac{B_o}{B_{ob}}$$

γ_o = Oil specific gravity =

$$141.5 / (API + 131.5)$$

T = Temperature, °F

□ For $P \geq P_b$:

$$B_o = B_{ob} \exp \left[\frac{C_o (P - P_b)}{1 + C_o (P - P_b)} \right] \quad (1.11)$$

where,

B_{ob} = Oil formation volume factor at the bubble point pressure

C_o = Oil compressibility, psi⁻¹

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c. Oil isothermal compressibility

The compressibility, or the bulk modulus of elasticity of a liquid, is defined by:

$$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (1.6)$$

Average compressibilities may be used by writing Eq. (1.6) in difference form as:

$$\beta = - \frac{1}{V} \ln \left(\frac{V_2}{V_1} \right) \quad (1.12)$$

The reference volume V in Eq. (1.31) may be V_1 , V_2 , or an average of V_1 , and V_2 .

□ Villena-Lanzi developed a correlation to estimate C_o for black oils. The correlation is good for pressures below the bubble-point pressure and is given by:

$$\ln C_o = -0.0001 \ln \left(\frac{P_b}{T} \right) - 0.0001 \ln \left(\frac{P_b}{T} \right) + 0.0001 \ln \left(\frac{P_b}{T} \right) + 0.0001 \ln \left(\frac{P_b}{T} \right) \quad (1.13)$$

where,

$$T = \text{°F}$$

The correlation was developed from a database containing the following ranges:

□ Vasquez and Beggs presented a correlation for estimating the compressibility for $P > P_b$.

This correlation is:

$$\mu = \mu_b \left(\frac{P}{P_b} \right)^{1.125} \quad (1.14)$$

The correlation gives good results for the following ranges of data:

Oil compressibility is measured from PVT laboratories and is used for:

□ Well inflow performance calculations

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□ Reservoir simulation

d. Oil viscosity

Viscosity (μ) is the resistance of fluid to flow.

Viscosity can be calculated by experimental correlations.

□ Egbogah's correlation for dead oil @ $P \leq P_b$ is: Figure 1.16. Oil viscosity.

$$\mu = \mu_b \left[\frac{P}{P_b} \right]^{0.16} \quad (1.15)$$

where,

μ_b = Dead oil viscosity, cp

T = Temperature, °F

The correlation was developed from a database containing the following ranges:

□ Beggs and Robinson developed the live oil viscosity correlation that is used in conjunction with the dead oil correlation given in Eq. (1.15)

$$\mu = \mu_b \quad (1.16)$$

where,

A = . +

B = . +

The oil samples contained the following ranges:

□ For $P > P_b$, the oil viscosity can be estimated by the following correlation developed by Vasquez and Beggs:

$$\mu = \mu_b \quad (1.17)$$

where,

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$$m = \mu_b \cdot \exp[-A - B] - \mu_b$$

μ_b = Oil viscosity at the bubble point pressure, cp

This correlation involved the following ranges:

Viscosity is used for:

□ Well inflow calculations.

□ Pressure drop calculations.

□ Effective mobility in multiphase flow.

Example 1.1. Use of correlations to estimate values for liquid properties at pressures of 2000 and 4000 psia.

Given:

= °F

= $\frac{5}{9}(T - 32)$

$\frac{5}{9} = .$

$\frac{5}{9}, \frac{5}{9} = ^\circ \frac{5}{9}$

$\frac{5}{9} = .$

Solution: Solution gas-oil ratio, R_{so} : $p = 4000\text{psia}$ ($p > p_b$)

For pressures greater than the bubble-point pressure, $R_{so} = R_{sob}$, therefore from Eq. (1.9):

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From Eq. (1.14):

From Eq. (1.13):

From Eq. (1.11):

$= \exp[\frac{p - p_b}{p_b}]$

R_{so} is calculated from Eq. (1.10):

$= . + . \frac{p - p_b}{p_b}$

From Eq. (1.10):

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From Eq. (1.17):

From Eq. (1.15):

From Eq. (1.16):

From Eq. (1.15), μ_{od} will be the same as μ_{obd} :

1.2.3 Review of gas properties

Gas properties can be calculated by theoretical methods beside experimental correlations or PVT laboratory measurement.

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Properties that will be discussed in this section are:

- ☐ Gas specific gravity
- ☐ Gas formation volume factor and density
- ☐ Gas isothermal compressibility
- ☐ Gas viscosity

a. Ideal gas law

Relationships that describe the pressure volume temperature (PVT) behavior of gases are called equations of state. The simplest equation of state is called the ideal gas law and is given by:

$$PV = nRT \quad (1.18)$$

where,

P = Absolute pressure

V = Total volume

n = Moles

T = Absolute temperature

R = Gas constant

When $R = 10.73$, p must be in pounds per square inch absolute (psia), V in cubic feet, n in pound-moles (lb-moles), and T in degrees Rankine. The ideal gas law was developed from Boyle's and Charles's laws, which were formed from experimental observations.

In petroleum industry, the standard conditions are usually 14.7 psia and 60 °F, the volume of gas at these conditions is given the units of SCF (standard cubic feet).

Example 1.2. Calculating the contents of a tank of ethane in moles, pounds, molecules, and SCF.

Given: A 500 ft³ tank of ethane at 100 psia and 100 °F.

Solution: Assuming ideal gas behavior:

$$pV = nRT$$

$$n = \frac{pV}{RT}$$

$$n = \frac{100 \text{ psia} \times 500 \text{ ft}^3}{10.73 \text{ (psia-ft}^3\text{)/(lb-mole-R)} \times 100 \text{ R}}$$

At 14.7 psia and 60°F:

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$$n = \frac{pV}{RT}$$

Alternate solution using Eq. (1.18):

$$pV = nRT$$

$$n = \frac{pV}{RT}$$

b. Gas specific gravity

Gas specific gravity is defined as the ratio of a gas at a given temperature and pressure to the density of air at the same temperature and pressure.

$$\text{density} = \frac{\rho_{\text{gas}}}{\rho_{\text{air}}} \quad (1.19)$$

$$\rho_{\text{gas}} = \frac{pM}{RT}$$

$$\text{density} = \frac{pM}{RT} = \frac{pM_{\text{air}}}{RT} \quad (1.20)$$

□ The molecular weight of air is usually taken as equal to 28.97 (79% nitrogen and 21% oxygen).

$$* \quad$$

$$\frac{M_{\text{gas}}}{M_{\text{air}}} = (1.21)$$

$$\frac{M_{\text{gas}}}{M_{\text{air}}} =$$

$$\frac{M_{\text{gas}}}{M_{\text{air}}} = \frac{pM_{\text{gas}}}{pM_{\text{air}}} = (1.22)$$

$$\frac{M_{\text{gas}}}{M_{\text{air}}} =$$

where,

M_w = molecular weight

c. Real gas law

All that has been said applies to a perfect or ideal gas. Actually there are no perfect gases; however,

many gases near atmospheric temperature and pressure approach ideal behavior.

All molecules of real gases have two tendencies:

□ To fly apart from each other because of their constant kinetic motion.

□ To come together because of electrical attractive forces between the molecules.

Because the molecules are quite far apart, the attractive forces are negligible, and the gas behaves close to ideal.

The supercompressibility factor (z), which is a measure of the amount the gas deviates from perfect behavior, is also called the gas deviation factor or z factor.

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$$z = (1.23)$$



□ This dimensionless quantity varies usually between 0.70 and 1.20.

□ A value of 1.00 representing ideal behavior.

Equation (1.23) may be substituted in the ideal gas law, Eq. (1.18) to give an equation for use with non-ideal gases

$$P = \frac{z n R T}{V} \quad (1.24)$$

□ The deviation factor of natural gas is commonly measured in the laboratory on samples of surface gases.

Figure 1.17. Effect of pressure, temperature, and composition on the gas deviation factor.

□ The gas deviation factor is commonly determined by measuring the volume of a sample at desired pressures and temperatures, and then measuring the volume of the same quantity of gas at atmospheric pressure and at a temperature sufficiently high so that all the material remains in the vapor phase.

Example 1.3. A sample of the Bell Field gas has a measured volume of 364.6 cm³ at 213 °F and 3250 psia. At 14.80 psia and 82°F it has a volume of 70,860 cm³. Calculating the gas deviation factor at 3250 psia and 213 °F.

Solution:

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$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$z_1 = \frac{P_1 V_1}{n R T_1}$$

$$z_2 = \frac{P_2 V_2}{n R T_2}$$

If the gas deviation factor is not measured, it may be estimated from its specific gravity. Example 1.4 below shows the method for estimating the gas deviation factor from its specific gravity.

The correlation to estimate pseudocritical temperature and pressure values for a gas with a given specific gravity was developed by Sutton, shown in Fig. 1.18. The range of specific gas gravities he worked with was $0.57 < \text{SG} < 1.68$:

$$T_{pc} = 170 + 320 \text{SG} - 130 \text{SG}^2 \quad (1.25)$$

$$P_{pc} = 70.6 + 5.9 \text{SG} - 0.1 \text{SG}^2 \quad (1.26)$$

Figure 1.18. Pseudocritical properties of natural gases.

Example 1.4. Calculating the gas deviation factor of the Bell Field gas from its specific gravity.

Given:

Specific gravity = 0.665

Reservoir temperature = 213 °F

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Reservoir pressure = 3250 psia

Figure 1.19. Compressibility factors for natural gases.

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Solution:

From Fig. 1.18, the pseudocritical pressure and temperature are

Using Eq. (1.25) and (1.26), the pseudocritical values are

Enter Fig. 1.19 with these values. Find $z = 0.918$.

A more accurate estimation of the deviation factor can be made when the analysis of the gas is available. This calculation assumes that each component contributes to the pseudocritical pressure and temperature in proportion to its volume percentage in the analysis and to the critical pressure and temperature, respectively, of that component.

Table 1.1 gives the critical pressures and temperatures of the hydrocarbon compounds and others commonly found in natural gases. It also gives some additional physical properties of these compounds. Example 1.5 shows the method of calculating the gas deviation factor from the composition of the gas.

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Table 1.1 Physical properties of the paraffin hydrocarbons and other compounds

Example 1.5. Calculating the gas deviation factor of the Bell Field gas from its composition.

Given: The composition Col. (2), and the physical data Cols. (3) to (5) taken from Table 1.1.

Solution:

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The specific gravity may be obtained from the sum of Col. (6), which is the average molecular weight of the gas,

$\rho_g = \frac{M_g}{M_{air}}$

The sums of Cols. (7) and (8) are the pseudocritical pressure and temperature, respectively. Then at 3250 psia and 213 °F, the pseudoreduced pressure and temperature are

$P_{pr} = \frac{P}{P_{pc}}$

$T_{pr} = \frac{T}{T_{pc}}$

The gas deviation factor using Figure 1.19 is $z = 0.91$.

d. Gas formation volume factor and density

Gas formation volume factor (B_g) related to the volume of gas in the reservoir to the volume on the surface, that is:

$B_g = \frac{V_{res}}{V_{std}}$ (1.27)

Where P_{sc} is 14.7 psia and T_{sc} is 60 °F,

$\rho_g = \frac{P}{P_{sc}} \frac{T_{sc}}{T} \frac{Z_{sc}}{Z}$ ft³/SCF = . bbl/SCF (1.28)

The constants in Eq. (1.28) are for 14.7 psia and 60 °F only, and different constants must be calculated for other standards.

Because gas is compressible, its density depends on pressure and temperature. Gas density can be calculated from gas law for real gas with good accuracy by Eq. (1.22). The moles of gas in 1 ft³ of reservoir gas pore space is P/zRT , therefore, the molecular weight of a gas is 28.97 lb per mole.

The reservoir gas density is:

$$\rho_g = \frac{P}{zRT} \times M$$

$$\rho_g = (1.29) \frac{P}{zT} \quad (1.29)$$

Example 1.6. Calculate the density of the Bell Field reservoir gas with a gas gravity of 0.665.

Solution:

$$\rho_g = \frac{P}{zRT} \times M$$

$$\rho_g = \frac{P}{zRT} \times M \quad \text{lb/cu ft}$$

$$\rho_g = \frac{P}{zRT} \times M$$

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e. Gas isothermal compressibility

Like oil, compressibility factor for gas is defined as the change in volume with pressure for gases under isothermal conditions, which is closely realized in reservoir gas flow, expressed by the real gas law:

$$V = \frac{zRT}{P} \quad (1.30)$$

Because

$$\beta = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (1.31)$$

Example 1.7. For an ideal gas $z = 1.00$ and $dz/dP = 0$ and the compressibility is simply the reciprocal of the pressure. An ideal gas at 1000 psia, then, has a compressibility of $1/1000$ or $1000 \times 10^{-6} \text{ psi}^{-1}$. Find the compressibility of a gas from the gas deviation factor curve of Fig. 1.19 using Eq. (1.31).

Given: The gas deviation factor curve for a gas at 150 °F, Fig. 1.20.

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Figure 1.20. Gas compressibility from the gas deviation factor versus pressure plot.

Solution:

At 1000 psia, the slope dz/dP is shown graphically in Fig. 1.19 as -0.0001 , Note that this is a negative slope. Then, because $z = 0.83$:

At 4500 psia the slope dz/dP is positive and as shown in Fig. 1.19 is equal to 0.0001 psi⁻¹. Since $z = 0.90$ at 4500 psia:

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Trube has replaced the pressure in Eq. (1.31) by the product of the pseudo-critical and the pseudo-reduced pressures, or $P = P_{pc}(P_{pr})$ and $dP = P_{pc}dP_{pr}$.

This obtains

$$\beta = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (1.32)$$

Multiplying through by the pseudo-critical pressure, the product $C_g(P_{pc})$ is obtained, which

Trube defined as the pseudo-reduced compressibility, $C_{r,}$ or

(1.33)

Mattar, Brar, and Aziz developed an analytical expression for calculating the pseudo-reduced compressibility. The expression is

(1.34)

Dranchuk and Abou-Kassem obtained:

(1.35)

and

(1.36)

Using Eqs. (1.34) to (1.36) and the definition of the pseudo-reduced compressibility, the gas compressibility can be calculated for any gas as long as the gas pressure and temperature are within the ranges specified for the Dranchuk and Abou-Kassem correlation.

f. Gas viscosity

The viscosity of natural gas depends on the temperature, pressure, and composition of the gas. It is not commonly measured in the laboratory because it can be estimated with good precision. Carr,

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Kobayashi, and Burrows have developed correlation charts, Fig. 1.20 and 1.21, for estimating the viscosity of natural gas from the pseudo-reduced critical temperature and pressure.

Figure 1.21. The viscosity of hydrocarbon gases at one atmosphere and reservoir temperatures, with corrections for nitrogen,

carbon dioxide, and hydrogen sulfide. (After Carr, Kobayashi, and Burrows, Trans. AIME.

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Figure 1.22. Viscosity ratio as a function of pseudo-reduced temperature and pressure. (After Carr, Kobayashi, and Burrows,

Trans. AIME.

□ The viscosity at one atmosphere and reservoir temperature is multiplied by the viscosity ratio to obtain the viscosity at reservoir temperature and pressure.

Example 1.8. Use of correlation charts to estimate reservoir gas viscosity.

Given:

Reservoir pressure = 2680 psia

Reservoir temperature = 212 °F

Well fluid specific gravity = 0.90 (Air = 1.00)

Pseudocritical temperature = 420 °R

Pseudocritical pressure = 670 psia

Carbon dioxide content = 5 mole %

Solution:

$\mu_1 = 0.0117$ cp at one atmosphere (Fig. 1.20)

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Correction for CO₂ = 0.0003 cp (Fig. 1.20, insert)

$\mu/\mu_1 = 1.60$ (Fig. 1.21)

Lee, Gonzalez, and Eakin developed a semiempirical method that gives an accurate estimate of gas viscosity for most natural gases if the z factor has been calculated to include the effect of contaminants. The ranges of variables used in the correlation were $100 < P(\text{psia}) < 8000$, $100 < T(^{\circ}\text{F}) < 340$, and $0.90 < \text{CO}_2 (\text{mole } \%) < 3.20$.

In addition to the gas temperature and pressure, the method requires the z factor and molecular weight of the gas. The following equations are used in the calculation for the gas viscosity:

(1.37)

where,

(1.38)

(1.39)

(1.40)

(1.41)

where,

ρ = Gas density, g/cc

P = Pressure, psia

T = Temperature, $^{\circ}\text{R}$

Mw = Gas molecular weight

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1.2.4 Review of water properties

Correlating the physical properties of reservoir water is relatively simple because, in general, the composition of the water is affected only by dissolved solids. Also, changes in the physical properties of water temperature and pressure are relatively small and can generally be predicted.

a. Solution gas-water ratio

McCain developed a correlation for the solution gas-water ratio, R_{sw} (SCF/STB). The correlation is:

$\diamond (- . - .)$

= (1.42)

(1.43)

□ The range of data for Eq. (1.42) was:

□ The range of data for Eq. (1.43) was:

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b. Water formation volume factor

McCain has also developed the following correlation for the water formation volume factor, B_w (bbl/STB):

= $+\Delta +\Delta$ (1.44)

where,

c. Water isothermal compressibility

Osif developed a correlation for the water isothermal compressibility, C_w , for pressures greater than the bubble point pressure. The equation is

(1.45)

□ The correlation was developed for the following range of data:

The water isothermal compressibility has been found to be strongly affected by the presence of free gas. Therefore, McCain proposed using the following expression for estimating C_w for pressures below or equal to the bubble point pressure:

(1.46)

□ The first term on the right-hand side of Eq. (1.46) is simply the expression for C_w in Eq. (1.45).

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□ The second term on the right-hand side is found by differentiating Eq. (1.43) with respect to pressure, or:

where,

B and C are defined in Eq. (1.43).

□ Eq. (1.46) should be used only for gross estimations of C_w .

d. Water viscosity

McCain developed the following correlation for water viscosity at atmospheric pressure and reservoir temperature:

◆ = (1.47)

□ The viscosity of water increases with decreasing temperature and in generally with increasing pressure and salinity.

□ Pressure below about 70°F causes a reduction in viscosity, and some salts (e.g., KCl) reduce the viscosity at some concentrations and within some temperature ranges.

□ The effect of dissolved gases is believed to cause a minor reduction in viscosity.

□ The range for Eq. (1.47) was:

The water viscosity can be adjusted to reservoir pressure by the following correlations, again developed by McCain:

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(1.48)

□ This correlation was found to be accurate to within 4% for pressures below 10,000 psia and within 7% for pressures between 10,000 and 15,000 psia. The temperature was between 86 and 167°F

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